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Inorganic Chemistry

ENTHUSIAST COURSE



STUDY MATERIAL

Metallurgy

English Medium

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METALLURGY

❑ INTRODUCTION :

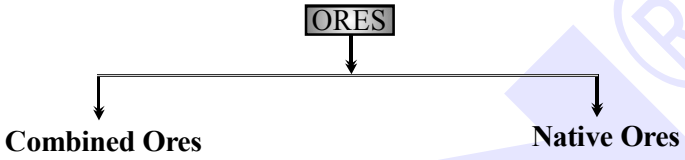
The compound of a metal found in nature is called a **mineral**.

The minerals from which metal can be economically and conveniently extracted are called **ores**.

An ore is usually contaminated with earthy or undesired materials known as **gangue**.

Note : All minerals are not ores but all ores are minerals.

Ores may be classified mainly into following two classes.



(a) ***Native ores :*** Silver, gold, platinum etc, occur as native ores.

(b) ***Combined ores :*** They contain the metal in combined form.

(i) ***Oxidised ores :*** Oxide ores , Carbonate ores, Sulphate ores, Phosphate ores, Silicate ores.

(ii) ***Sulphurised ores :*** These ores consist of sulphides of metals like iron, lead, zinc, mercury etc.

(iii) ***Halide ores :*** These ores consist of halides of metals

❑ METALLURGY :

The scientific and technological process used for the extraction/isolation of the metal from its ore is called as metallurgy.

The isolation and extraction of metals from their ores involve the following major steps:

(A) Crushing of the ore.

(B) Dressing or concentration of the ore.

(C) Isolation of the crude metal from its ore

(D) Purification or refining of the metal.

(A) Crushing and Grinding : The ore is first crushed by crushers and ground to a powder.

(B) Concentration of the ore : The removal of unwanted, useless impurities from the ore is called ***dressing, concentration or benefaction of ore.***

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There are several steps for the concentration of Ores.

(I) By physical separation

(a) Gravity separation (Levigation)

(b) Froth Floatation method

(c) Magnetic separation

(i) Gravity separation or Levigation method :

It is based on the difference in the densities of the gangue and ore particles. This method is generally used for the concentration of oxide and native ores.

(ii) Electromagnetic separation :

It is based on differences in magnetic properties of the ore components.

It is used when either the ore or the impurities associated with it are magnetic in nature.

Examples :

Chromite ore($FeO.Cr_2O_3$) is separated from non–magnetic silicious impurities and cassiterite ore(SnO_2) is separated from magnetic Wolframite ($FeWO_4 + MnWO_4$).

(iii) Froth floatation process : This process is based on differential wetting of the ore by oil and gangue by water.

Examples : Galena, PbS (ore of Pb) ; copper pyrites $Cu_2S.Fe_2S_3$ or $CuFeS_2$ (ore of copper) ; zinc blende, ZnS (ore of zinc) etc.

(a) Frothers : Oil like pine oil, camphor oil etc., are used as frothers.

(b) Frothers stablizer : Aneline & Cressol

(c) Collectors : Potassium or sodium ethyl xanthate is used as a collector.

(d) Activating and depressing agents : For example galena (PbS) usually contains the minerals namely zinc blende (ZnS) and pyrites (FeS_2) as impurities. Floatation is carried out by using potassium ethyl xanthate (used as a collector) along with NaCN and Na_2CO_3 (used as depressing agent).

(II) By Chemical separation

Some of the ores are concentrated by means of chemical treatment.

Leaching : It involves the treatment of the ore with a suitable reagent. as to make it soluble while impurity remain insoluble. The ore is recovered from the solution by suitable chemical method.

(i) Bayer's process

(ii) Cyanide process

❑ CALCINATION

Calcination is a process in which ore is heated, generally in the **absence of air**, to expel water from a hydrated oxide or carbon dioxide from a carbonate at temperature below their melting points.

For Example

(i) $CaCO_3 \longrightarrow CaO + CO_2$

(ii) $Al_2O_3.\ 2H_2O \longrightarrow Al_2O_3 + 2H_2O$

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❑ ROASTING

The removal of the excess sulphur contained in sulphide ores by heating **in an excess of air** is called roasting.

(Metal sulphides $\xrightarrow{+O_2}$ Metal oxide + SO_2)

(iii) $2Cu_2S + 3O_2 \longrightarrow 2Cu_2O + SO_2\uparrow$

❑ THERMODYNAMICS OF REDUCTION PROCESSES (ELLINGHAM DIAGRAM)

The extraction of metals from their oxides using carbon or other metals, and by thermal decomposition, involves a number of points which merit detailed discussion.

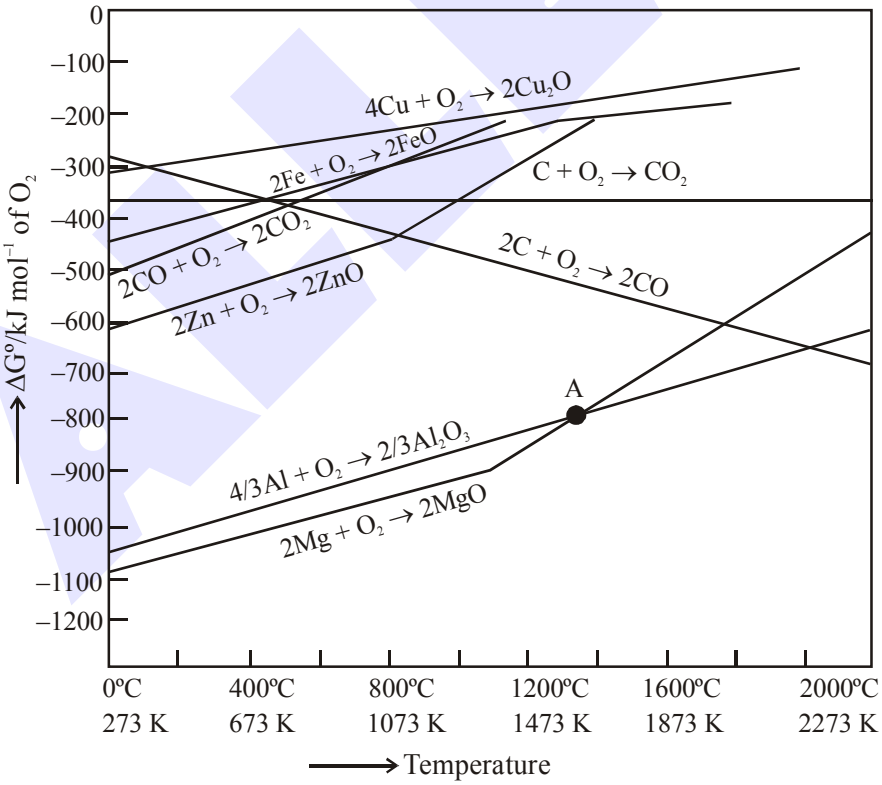
For a spontaneous reaction, the free energy change ΔG must be negative.

$$\Delta G = \Delta H - T\Delta S$$

ΔH is the enthalpy change during the reaction, T is the absolute temperature, and ΔS is the change in entropy during the reaction. Consider a reaction such as the formation of an oxide:

$$M + O_2 \rightarrow MO$$

Dioxygen is used up in the course of this reaction. Gases have a more random structure (less ordered) than liquids or solids. Consequently gases have a higher entropy than liquids or solids. In this reaction S the entropy or randomness decreases, the hence ΔS is negative. Thus if the temperature is raised then $T\Delta S$ becomes more negative. Since $T\Delta S$ is subtracted in the equation, then ΔG becomes less negative. Thus the free energy changed increases with an increase of temperature.



The free energy changes that occur when one gram molecule of a common reactant (in this case dioxygen) is used may be plotted graphically against temperature for a number of reactions of metals of their oxides. This graph is shown in figure and is called an Ellingham diagram (for oxides). Similar diagrams can be produced for one gram molecule of sulphur, giving an Ellingham diagram for sulphides, and similarly for halides.

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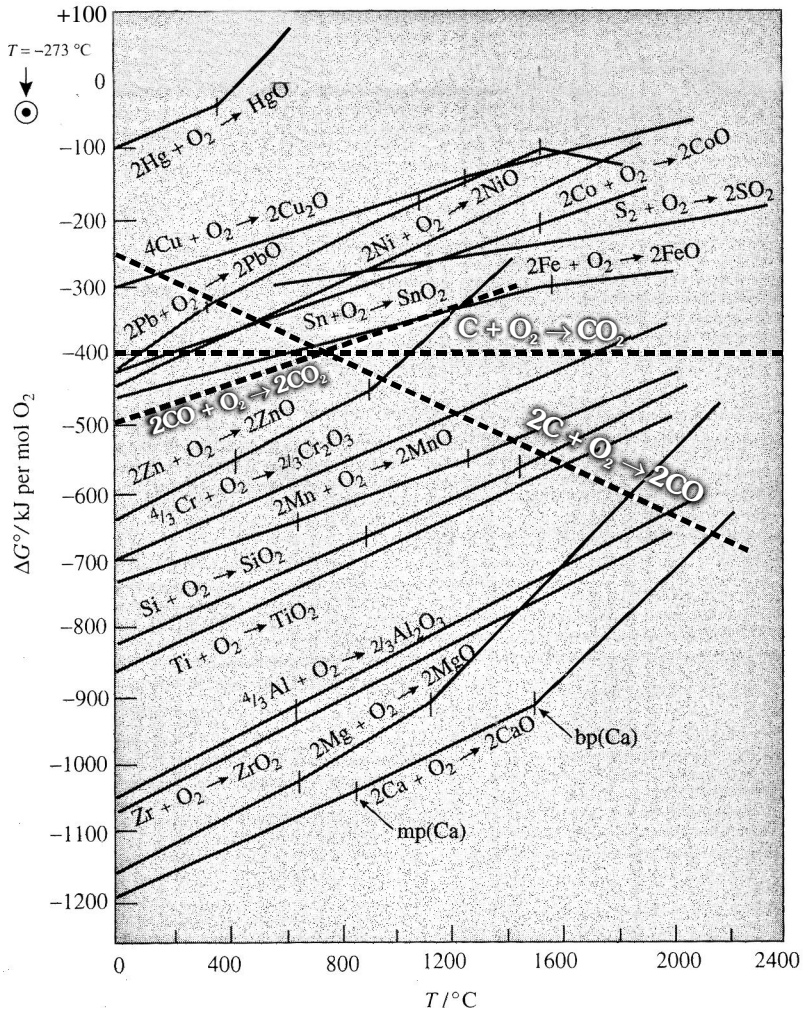
The Ellingham diagram for oxides shows several important features:

(i) The graph for metal oxide all slope upwards, because the free energy change increases with an increase of temperature as discussed above.

(ii) The free energy changes all follows a straight line unless the materials metal or vaporize.

(iii) When the temperature is raised, a point will be reached where the graph crosses the $\Delta G = 0$ line. Below this temperature the free energy of formation of the oxide is negative, so the oxide is stable. Above this temperature the free of formation of the oxide is positive, and the oxide becomes unstable, and should decompose into the metal and dioxygen.

(iv) Any metal will reduce the oxide of other metals which lie above it in the Ellingham diagram because the free energy will become more negative by an amount equal to the different between the two graphs at that particular temperature.



Limitations of Ellingham Diagram

(i) The graph simply indicates whether a reaction is possible or not i.e., the tendency of reduction with a reducing agent is indicated. This is so because it is based only on the thermodynamic concepts. It does not say about the kinetics of the reduction process (Cannot answer questions like how fast it could be ?).

(ii) The interpretation of ΔG^{Θ} is based on K($\Delta G^{\Theta} = -$ RT lnK). Thus it is presumed that the reactants of products are in equilibrium.

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❑ REDUCTION OF ORE TO THE METAL

The calcined or roasted ore is then reduced to the metallic state in either of the following ways.

(i) Reduction by Carbon (Smelting)

- Concentrate ore (ore + gangue) + R.A. (carbon) + Flux [$\therefore$ R.A. $\Rightarrow$ Reducing agent]

$\downarrow$

Metal + Slag + gases

$PbO + C \longrightarrow Pb + CO$

$Fe_2O_3 + 3C \longrightarrow Fe + 3CO$

Acidic impurity + Flux $\longrightarrow$ slag

$P_2O_5 + 3CaO \longrightarrow Ca_3(PO_4)_2$

Basic impurity + Flux $\longrightarrow$ slag

$MgCO_3 + SiO_2 \longrightarrow MgSiO_3 + CO_2 \uparrow$

$FeO + SiO_2 \longrightarrow FeSiO_3$

(ii) Self Reduction

Sulphides of certain metals are reduced to metal without using any additional reducing agent. ores of Cu, Pb, Hg etc.

Self Reduction for Pb

$2PbS + 3O_2 \xrightarrow{\text{Roasting}} 2PbO + 2SO_2 \uparrow$

(Galena) (air)

$PbS + 2PbO \xrightarrow[\text{Absence of air}]{\text{High temp}} 3Pb + SO_2 \uparrow$

(unroasted ore) (roasted ore) (Self reduction)

(iii) Metal Displacement Method

$2Na[Au(CN)_2] + Zn \longrightarrow Na_2[Zn(CN)_4] + 2Au$

$2Na[Ag(CN)_2] + Zn \rightarrow Na_2[Zn(CN)_4] + 2Ag$

Sodium tetra cyanozincate

(iv) Electrolytic Reduction

This process is mainly used for the extraction of **highly electropositive metals.**

Ex. Na, K, Mg, Ca, Al, etc.

(v) Thermite Reduction or Thermite Process

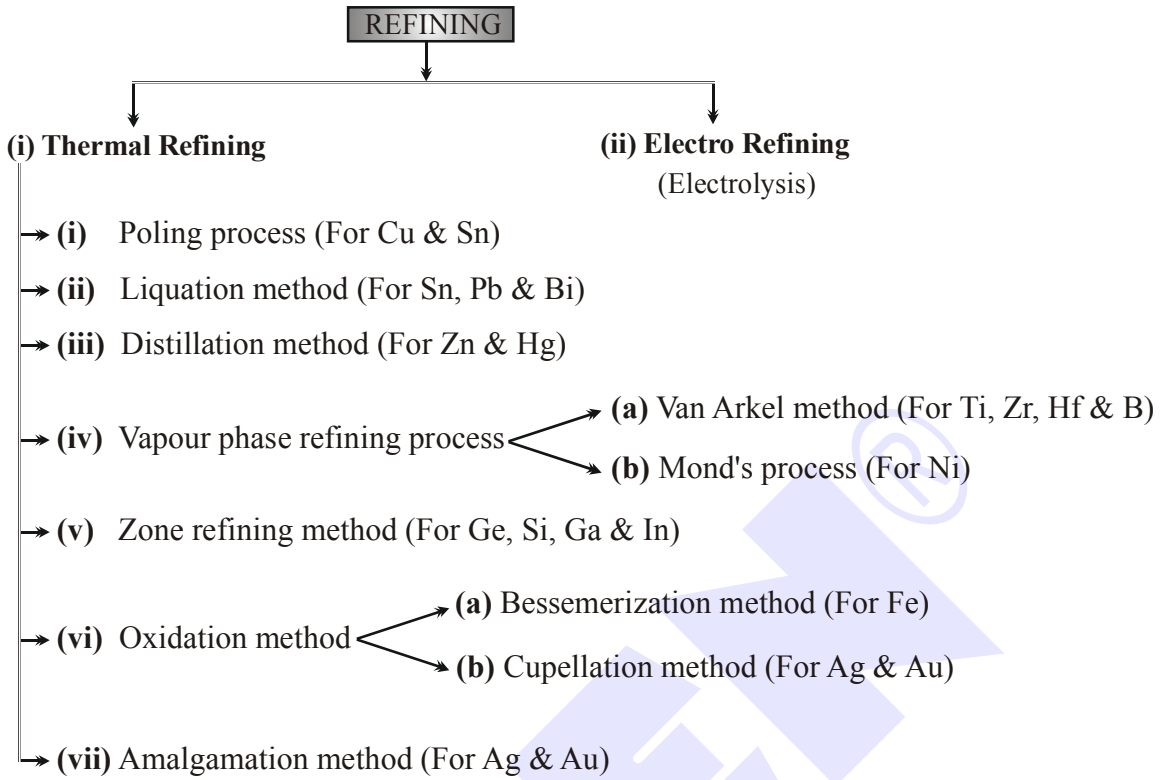
Al is used as reducing agent in this process. This process is employed in the case of those metals which have very high melting points and are to be extracted from their oxides

$Cr_2O_3 + 2Al \longrightarrow 2Cr + Al_2O_3$

$3Mn_3O_4 + 8Al \longrightarrow 9Mn + 4Al_2O_3$

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❑ REFINING OF METALS



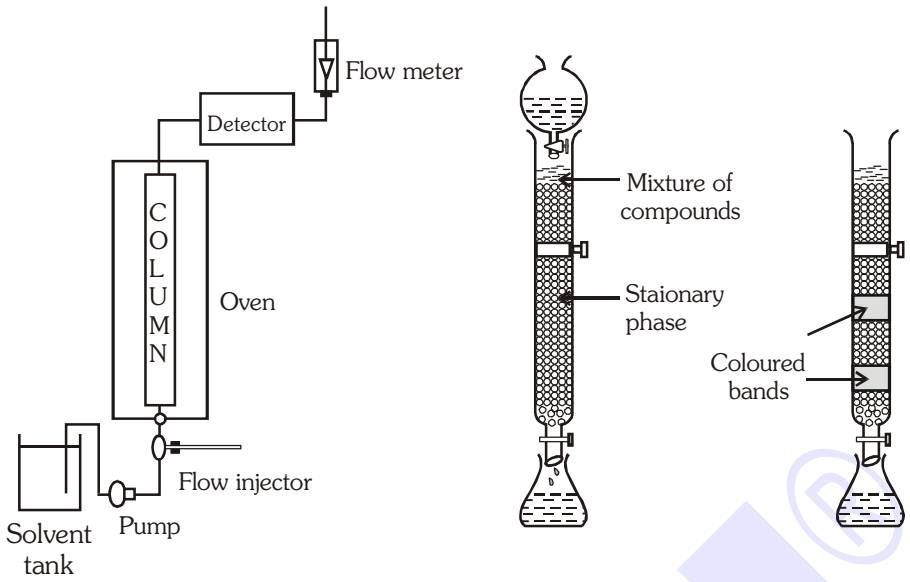
❑ OTHER METHOD : BY CHROMATOGRAPHIC METHOD

This method is based on the principle that different components of a mixture are differently adsorbed on an adsorbent. The mixture is put in a liquid or gaseous medium which is moved through the adsorbent.

Different components are adsorbed at different levels on the column. Later the adsorbed components are removed (eluted) by using suitable solvent (eluant). Depending upon the physical state of the moving medium and the adsorbent material and also on the process of passage of the moving medium, the chromatographic method is given the name. In one such method the column of Al_2O_3 is prepared in a glass tube and the moving medium containing a solution of the components is in liquid form.

This is an example of column chromatography. This is very useful for purification of the elements which are available in minute quantities and the impurities are not very different in chemical properties from the element to be purified. There are several chromatographic techniques such as paper chromatography, column chromatography, gas chromatography, etc. Procedures followed in column chromatography have been depicted in the following figures.

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(a) Industrial method **(b) Laboratory method**

Fig. Schematic diagrams showing column chromatography

EXTRACTION OF SOME INDIVIDUAL METALS

❑ EXTRACTION OF COPPER

(1) Extraction of copper :

Occurrence : Copper occurs in free as well as in combined state. The main ores are as follows.

(i) Copper pyrites $CuFeS_2$ or $Cu_2S.Fe_2S_3$

(ii) Cuprite (Ruby copper) Cu_2O

(iii) Copper glance Cu_2S

(iv) Malachite $Cu(OH)_2.CuCO_3$

(v) Azurite $2Cu(OH)_2.CuCO_3$

Extraction :

Copper is extracted from the ores (sulphide) by two process.

(A) Pyrometallurgical Process :

This is a dry process and applied to high grade ores (containing 4% or more copper).

(B) Hydrometallurgical Process :

This is a wet process and applied to low grade ores.

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(A) Extraction from pyrites by pyrometallurgical process (Smelting Process)

(i) Concentration :

The finely powdered ore is concentrated by froth floatation process.

(ii) Roasting :

The concentrated ore is heated strongly in presence of air.

Sulphur, arsenic, and antimony are removed in form of their volatile oxides while the ore is converted into a mixture of cuprous and ferrous sulphides.

These sulphides are partially oxidised to oxides.

(a) $S + O_2 \rightarrow SO_2 \uparrow$

(b) $2As_2S_3 + 9O_2 \rightarrow 2As_2O_3\uparrow + 6SO_2 \uparrow$

(c) $2Sb_2S_3 + 9O_2 \rightarrow 2Sb_2O_3 \uparrow + 6SO_2 \uparrow$

(d) $2CuFeS_2 + O_2 \rightarrow Cu_2S + 2FeS + SO_2$

$2FeS + 3O_2 \rightarrow 2FeO + 2SO_2$

(iii) Smelting

The roasted ore is mixed with sand (flux) and coke (fuel) and then heated in a reverberatory furnace water jacketed blast furnace called smelter. The oxidation of ferrous sulphide which started during roasting now goes a step further. Ferrous oxide formed, reacts with sand to form ferrous silicate (slag).

Here some FeS reacts with Cu_2O to form Cu_2S again.

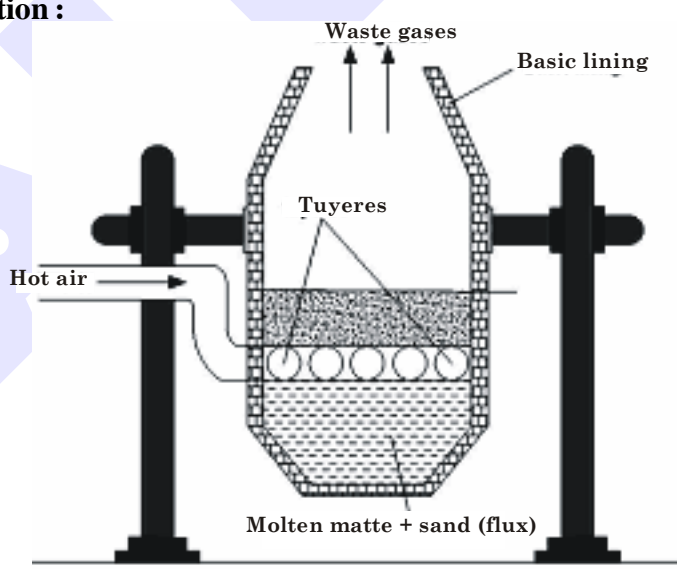
(a) $FeO + SiO_2 \rightarrow FeSiO_3$ (Slag)

(b) $2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$

(c) $Cu_2O + FeS \rightarrow Cu_2S + FeO$

Slag (forming upper layer) and molten mass consisting of Cu_2S and a little FeS **(forming lower layer and commonly called matte)** are removed from separate holes.

(iv) Bessemerisation :



Molten matte is heated in a Bessemer converter and a blast of air mixed with sand is blown through the molten mass.

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Here iron (FeS) is completely removed as slag, a part of Cu_2S is oxidised to Cu_2O and a part of Cu_2S reduces Cu_2O to metallic copper.

(a) $2FeS + 3O_2 \rightarrow 2FeO + 2SO_2 \uparrow$

$FeO + SiO_2 \rightarrow FeSiO_3$

(b) $2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2 \uparrow$

$2Cu_2O + Cu_2S \rightarrow 6Cu + SO_2\uparrow$

The molten copper is poured off into moulds. As it cools it gives up the dissolved sulphur dioxide which forms blister on the surface of the metal. Hence the metal thus obtained is called blister copper and contains 98 percent copper.

(v) Refining of copper :

(a) Poling :

The molten blister copper is heated in pressure of air and stirred with green wood **poles**. S and As are oxidised to their volatile oxides.

Iron is oxidised and forms a scum or slag which is **skimmed** off.

The cuprous oxide is reduced to Cu by CO and CH_4 from green wood.

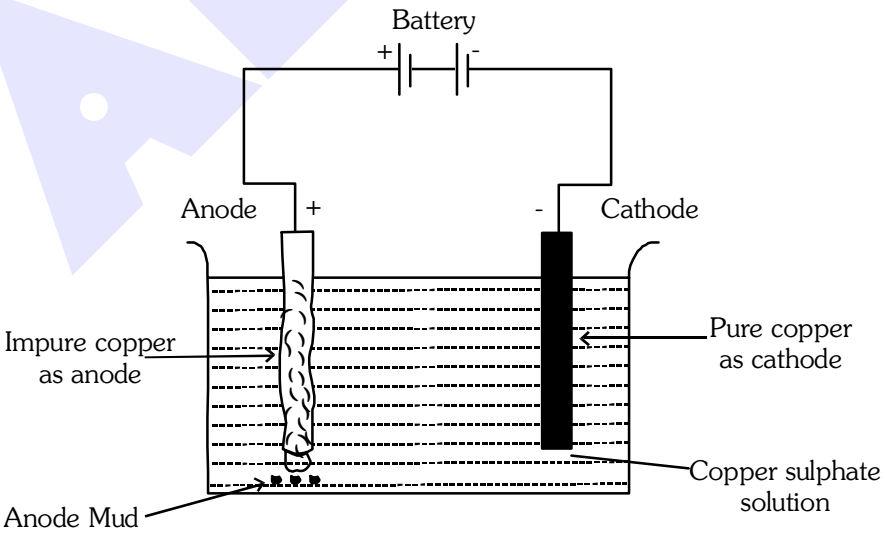
$Cu_2O + CO \rightarrow 2Cu + CO_2\uparrow$

$4Cu_2O + CH_4 \rightarrow 8Cu + 2H_2O + CO_2$

(b) Electrolytic refining :

The electrolytic bath contains an acidified solution of copper sulphate. Impure copper acts as anode while cathode is of the pure copper strip. When electric current is passed, there is transfer of pure copper from anode to cathode which gradually grows in size.

The impurities like Fe, Zn, Ni, Co etc. dissolve in the solution as sulphates and others like Au and Ag settle down below the anode as **anode mud.** The cathode is removed and copper of about 99.99% purity is obtained.



Electrolytic refining of copper

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(B) Hydrometallurgical Process :

Crushed and powdered low grade ore is exposed to air and water then copper sulphide is oxidised to copper sulphate. Some iron sulphate and sulphuric acid is also produced. The process is completed in about a year.

(a) $CuFeS_2 + 4O_2 \rightarrow CuSO_4 + FeSO_4$
(Copper pyrite)

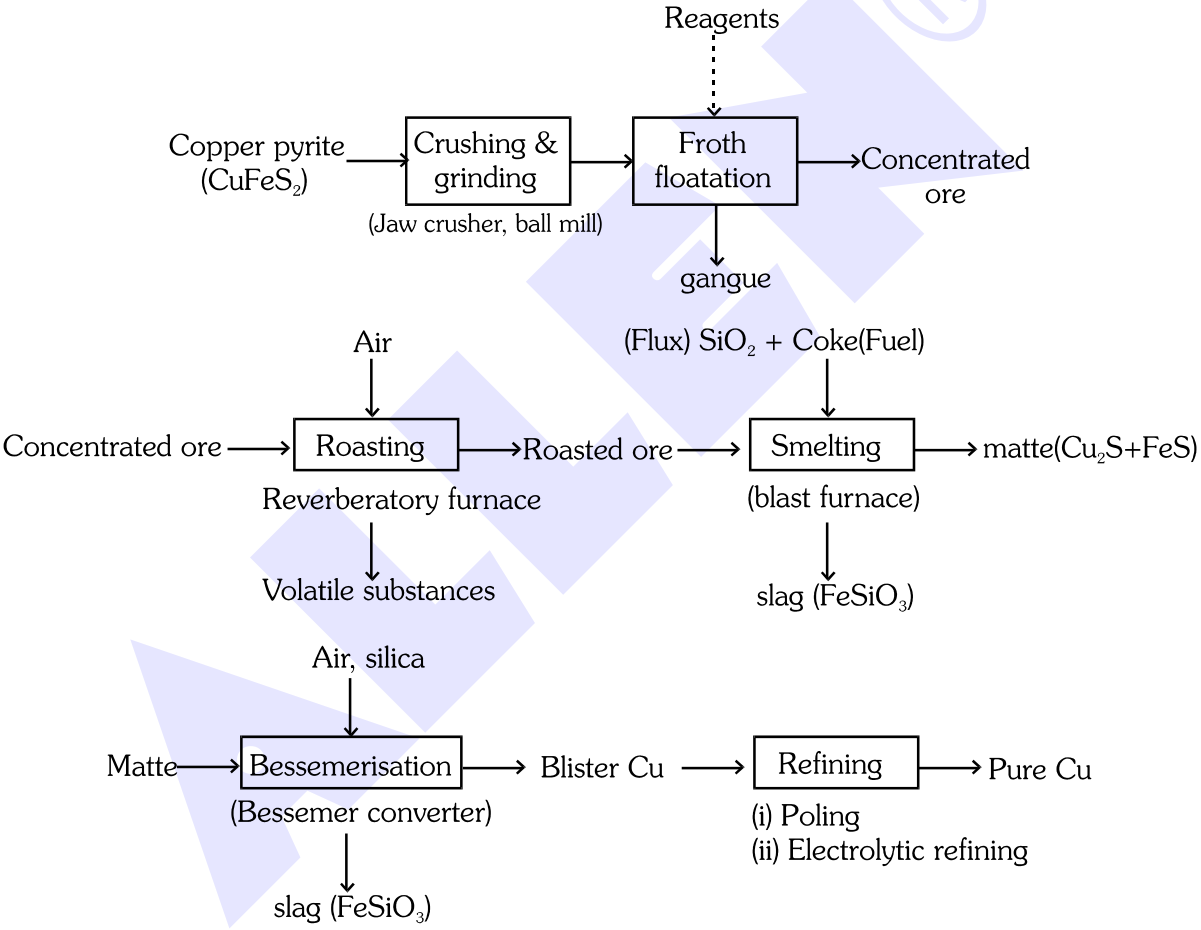
(b) $2Cu_2S + 5O_2 \rightarrow 2CuSO_4 + 2CuO$
(Copper glance)

(c) $CuCO_3.Cu(OH)_2 + 2H_2SO_4 \rightarrow 2CuSO_4 + CO_2 \uparrow + 3H_2O$
(malachite)

The pale green liquor draining from the bottom of the heaps is $CuSO_4$ which is collected in pans and treated with iron scraps which precipitate out copper.

$CuSO_4 + Fe/Zn/H_2 \rightarrow FeSO_4/ZnSO_4/H_2SO_4 + Cu$ (red ppt.) $\downarrow$

Process in brief :



Uses :

Copper is the second most useful metal (the first being Iron) because of its stability in air and water and excellent conductivity.

It is used :–

1. In the manufacture of electrical wires, cables etc.
2. For electroplating.
3. As a coinage metal and in ornaments and jewellery.
4. For the manufacture of alloys like brass (Cu + Zn), bronze (Cu + Sn) German silver (Cu + Zn + Ni) bell metal (Cu + Sn) gun metal (Cu + Sn + Zn), coppper coins (Cu + Zn + Sn) etc.

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EXTRACTION OF LEAD

(2) Extraction of lead :

Main Ore : Galena (PbS) : There are mainly two types of process used in the extraction of Lead.

Ore - **Galena PbS** → **(a) Carbon reduction process**
Galena PbS → **(b) Self reduction process**

(a) Carbon reduction process (When impurity content is high) :

(i) Crushing & Grinding

(ii) Concentration by Forth floatation method

(iii) Roasting process with Lime stone ($CaCO_3$). Roasting in air to give PbO and SO_2 .

$$2PbS(s) + 3O_2(g) \xrightarrow{\Delta} 2PbO(s) + 2SO_2$$

(iv) Smelting (Carbon reduction method with coke + Fe_2O_3). Then reduction with coke or CO in a blast furnace take place.

$$2PbO(s) + C \xrightarrow{\Delta} 2Pb(\ell) + CO_2(g)$$

(b) Self reduction process : (When the impurity content is less)

(i) Crushing & Grinding

(ii) Concentration by Forth floatation method

(iii) Self reduction process : PbS is partially oxidized by heating and blowing air through it. After some time the air is turned off and heating is continued. The mixture undergoes self reduction as given below.

$$PbS + O_2 \longrightarrow PbO + SO_2$$

$$PbS + 2PbO \longrightarrow 3Pb + SO_2$$

Parallel reaction

$$PbS + 2O_2 \longrightarrow PbSO_4$$

$$PbS + PbSO_4 \longrightarrow 2Pb + 2SO_2$$

(c) Refining process :

(a) Liquation

(b) Bett's electrorefining

Anode → Impure Pb

Cathode → Pure Pb

Electrolyte → $Pb[SiF_6] + H_2SiF_6$ + Gelatin (to adjust viscosity)

on the electrolysis Pb is deposited at cathode which give 99.95% pure metal.

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EXTRACTION OF ZINC

(2) Extraction of zinc :

Occurrence : Its important minerals are :-

(i) Zinc blende or black jack ZnS

(ii) Zincite ZnO

(iii) Calamine $ZnCO_3$

Extraction :

Electrolytic Process :

(i) Concentration :

The powdered ore ZnS is concentrated by froth floatation method.

(ii) (a) Roasting :

Concentrated ZnS is roasted at 700^0C. A mixture of ZnO and $ZnSO_4$ is obtained

$ZnS + 2O_2 \rightarrow ZnSO_4$

$2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$

$$ZnS \xrightarrow{\text{concentration}} ZnS \xrightarrow{\text{roasting}} ZnO + SO_2$$

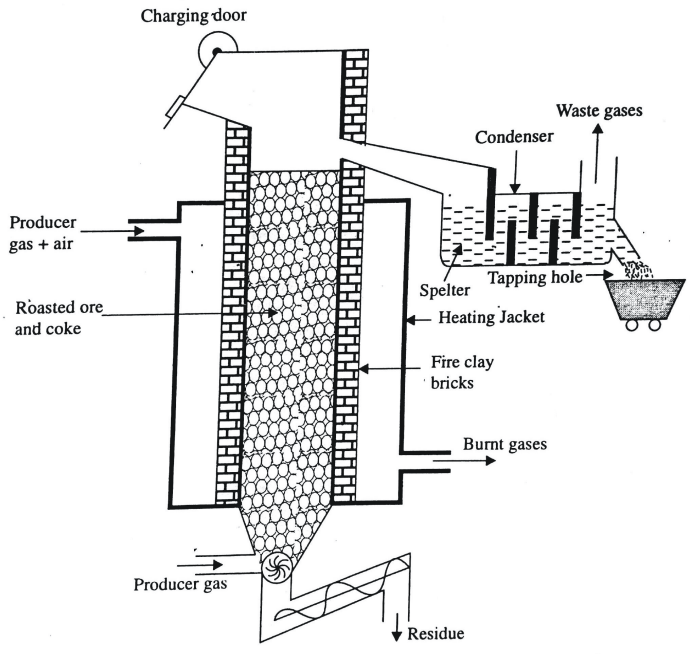
(b) Calcination Process

$$ZnCO_3 \xrightarrow{\text{concentration}} ZnCO_3 \xrightarrow{\text{calcination}} ZnO + CO_2$$

(iii) Carbon Reduction (Extraction of zinc from zinc oxide) :-

The reduction of zinc oxide is done using coke. The temperature in this case is higher than that in case of copper. For the purpose of heating, the oxide is made into brickettes with coke and clay.

$$ZnO + C \text{ (coke)} \xrightarrow{1673K} Zn + CO$$



Vertical retort process for the reduction of ZnO to Zn-metal.

The metal is distilled off and collected by rapid chilling is called **zinc spelter**.

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(iv) Electrolysis (Electrolytic refining) **:**

Anode → Impure Zn

Cathode → Thin Al-rod

Electrolyte → Solution of $ZnSO_4 + H_2SO_4$ (dil.)

on the electrolysis zinc is deposited at cathode. The metal is scrapped off and melted to give 99.95% pure metal.

Uses :

(i) In making alloys e.g. brass, german silver, elektron (Alloy of Mg with smaller amount of Al, Y, Ag, Gd, Zn) etc.

(ii) In the extraction of silver and gold by cyanide process.

(iii) It is also used in large quantities in batteries and dry cells for making cathode container.

(iv) Zn–Cu couple, Zn- Hg, zinc dust etc. are used as reducing agent in organic reactions.

(v) large amount of zinc is used for galvanizing iron. Zinc is deposited on the surface of iron articles. This process is called galvanization.

(vi) It is also used in large quantities in batteries, as a constituent of many alloys, e.g., brass, (Cu 60%, Zn 40%) and german silver (Cu 25-30%, Zn 25-30%, Ni 40–50%).

(vii) Zinc dust is used as a reducing agent in the manufacture of dye-stuffs, paints, etc.

EXTRACTION OF TIN

Extraction of tin from cassiterite :

Main Ore : Cassiterite or Tinstone (SnO_2) + Major impurities

[(i) SiO_2, (ii) Sulphides of Fe & Cu, (iii) $FeWO_4 + MnWO_4$]

Some following steps are involved :

(i) Crushing and concentration :

The ore is crushed and washed with a stream of running water to remove the lighter earthy and silicious impurities.

(ii) Electromagnetic separation :

The concentrated ore is subjected to the electromagnetic separation to remove magnetic impurity of Wolframite.

(iii) Roasting :

The ore is then heated in presence of air, when volatile impurities (S as SO_2, As as As_2O_3 and Sb as Sb_2O_3) are removed.The impurities of pyrites of copper and iron are converted into their respective oxides and sulphates

$CuS + 2O_2 \rightarrow CuSO_4$; $FeS + 2O_2 \rightarrow FeSO_4$

(iv) Leaching :

Sulphates of copper and iron are dissolved in water.

(v) Washing :

The ore is washed with running water to remove the fine iron oxide produced in roasting. Thus obtained ore contains 60 – 70% SnO_2 and is called as black tin.

(vi) Carbon reduction method : Coke & Lime stone (flux) is used.

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Smelting :

The black tin is mixed with anthracite coal and heated to about 1500K in a reverberatory furnace. If SiO_2 is present as impurity then CaO is added as flux.

$SnO_2 + C \rightarrow SnO + CO\uparrow$

$SnO + SiO_2 \rightarrow SnSiO_3$; $CaO + SiO_2 \rightarrow CaSiO_3$

$SnSiO_3 + CaO + C \rightarrow Sn + CaSiO_3 + CO\uparrow$

or use scrap iron $SnSiO_3 + Fe \rightarrow Sn + FeSiO_3$

Refining method :

(a) Poling (b) Electrorefining

Anode → Impure Sn

Cathode → Pure Sn

Electrolyte → ($SnSO_4$ solution + dil. H_2SO_4)

EXTRACTION OF IRON

(1) Extraction of Iron :

Main Ore : *Haematite (Fe_2O_3)*

Some following steps are involved :

(i) Crushing & Grinding :

(ii) Concentration : By gravity separation method.

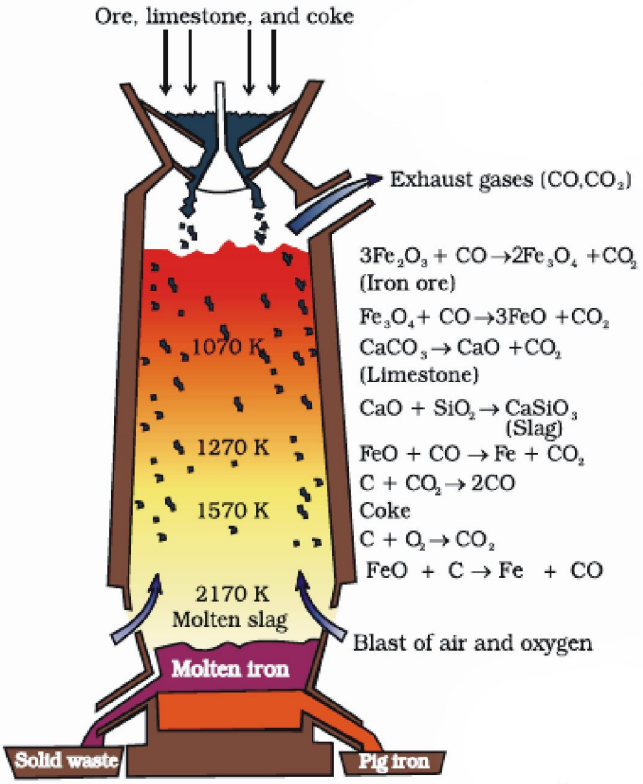
(iii) Roasting :

(iv) Carbon reduction (Blast furnace) : Pig iron is obtained from this process

Smelting or reduction :

The calcined ore (8 parts) mixed with coke, a reducing agent (4 parts) and lime stone a flux (1 part) is reduced in a blast furnace.

Since the blast furnace has different temperatures at different zones, different reactions take place at different zones.



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(a) Zone of combustion (1500 -1600^{0}C)

This zone is near at the bottom of the furnace and little above the tuyers. It increases temp. of the furnace because of exothermic reactions.

$2C + O_2 \rightarrow 2CO$ $\quad$ $C + O_2 \rightarrow CO_2$

(b) Zone of fusion (1200- 1300^{0}C)

It is just above the zone of combustion. Here the iron melts and trickles down in the hearth while the slag being lighter floats over the molten metal and thus prevents oxidation of Fe by blast of air.

(c) Zone of heat absorption or slag formation (800- 1000^{0}C)

This is the middle part of the furnace. Here rising CO_2 is reduced to carbon monoxide.

$CO_2 + C \rightarrow 2CO$

The reaction being endothermic, lowers the temperature of the zone.

Limestone decomposes forming CaO which reacts with SiO_2 forming slag.

$CaCO_3 \rightarrow CaO + CO_2$ $\quad$ $CaO + SiO_2 \rightarrow CaSiO_3$ (Slag)

(d) Zone of reduction (400-700^{0}C)

It is near the top of the furnace. Here the calcined ore is reduced to Fe by rising CO.

$Fe_2O_3 + CO \rightleftharpoons 2FeO + CO_2 \uparrow$

$Fe_3O_4 + CO \rightleftharpoons 3FeO + CO_2 \uparrow$

$FeO + CO \rightarrow Fe + CO_2 \uparrow$

To retard the backward reaction, supply of excess of CO is maintained by the following reaction.

$CO_2 + C \rightarrow 2CO$

Since the temperature of this zone is too low to melt iron, the metal produced is known as spongy iron.

Refining :

Purification of Fe can be done by different method which are as follows :

(a) Puddling Process

(b) Bessemerisation Process

(c) Open hearth Process

(d) L. D. Process

Thus we got pure iron.

Types of Iron

Cast iron or pig iron

It is most impure form of Iron and contains the highest proportion of carbon (2.5 - 4 %) along with traces of S, P, Mn and Si. Cast iron contain 2.5 to 4.3 & pig iron contain 2.5 to 5%.

Wrought iron (Fibrous iron) or malleable iron

It is the purest form of iron and contains minimum amount of carbon (0.12 - 0.25%) and less than 0.5% of other impurities.

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Steel

It is the most important form of iron and finds extensive applications. As far as carbon content (impurity) is concerned it is mid-way between cast iron and wrought iron, it contains 0.25- 2% carbon.

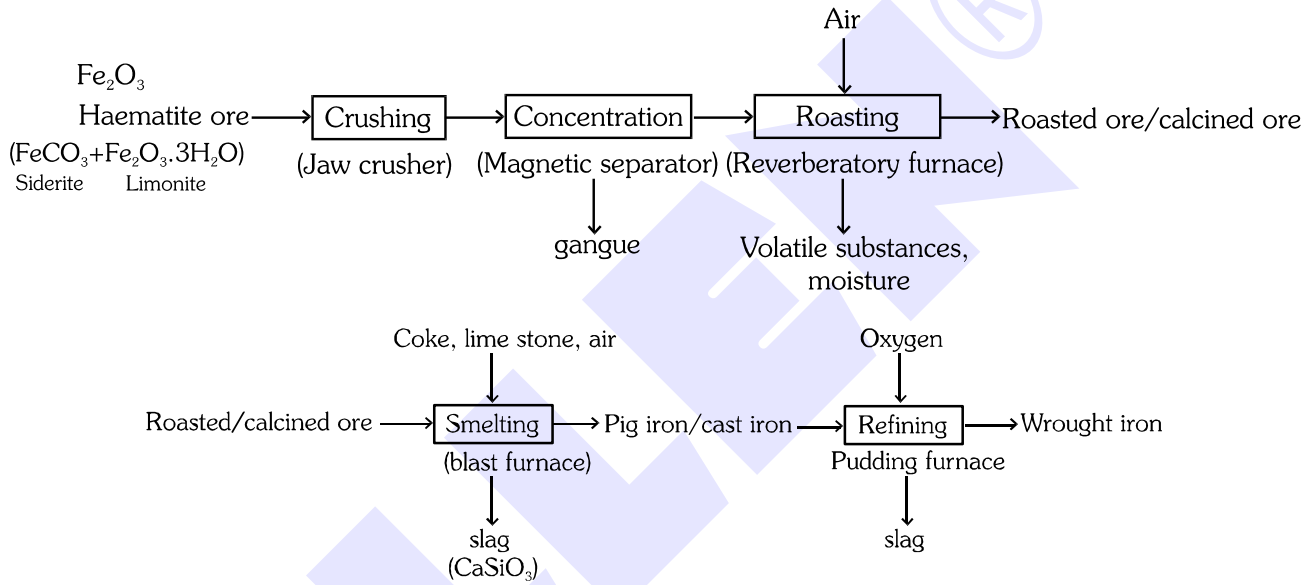
Thus all the three forms of iron differ in their carbon contents, both iron and steel are obtained from cast iron.

Order of M.P. Wrought Iron > Steel > Cast Iron or Pig Iron

% of Carbon in different type of Iron

	NAME	***% of C***
(1)	Wrought iron	0.1 to 0.25
(2)	Steel	0.25 to 2.0
(3)	Cast Iron	2.6 to 4.3
(4)	Pig Iron	2.3 to 4.6

Process in brief :



Manufacture of Steel : The addition of different desired impurities into molten pure iron is known as steel making

Bessemer process :

This process involves the use of a large pear- shaped furnace (vessels) called Bessemer converter. This is made of steel plates lined with silica (SiO_2) or magnesia (MgO) depending upon the nature of impurities. If the impurities are acidic e.g. P_4O_{10} or SiO_2, basic lining of lime (CaO) or magnesia is used (Basic process). If the impurities are basic e.g. MnO, lining of silica bricks is used (acid process). Silicon and manganese (in acidic process) or phosphorus (in basic process) are oxidised to their oxides and thus removed as slag.

Acid process : $Si + O_2 \rightarrow SiO_2$; $2Mn + O_2 \rightarrow 2MnO$; $MnO + SiO_2 \rightarrow MnSiO_3$ (Slag)

Basic process : $P_4 + 5O_2 \rightarrow P_4O_{10}$ $P_4O_{10} + 6CaO \rightarrow 2Ca_3(PO_4)_2$ (Thomas slag)

Properties of Steel :

Steel combines the useful properties of cast iron and wrought iron. It is hard and elastic. The properties of steel depend upon its carbon content. With the increase in carbon content, the hardness of steel increases while its ductility decreases.

(i) Low carbon or soft steel contain C upto 0.25%.

(ii) Medium carbon steel or mild steel contain 0.25–0.5% C.

(iii) High carbon or hard steel contain 0.5–1.5% C.

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Heat Treatment of Steel

(i) ***Quenching or hardening :*** Steel is heated to red hot temperature (700 to 800°C) and is then cooled suddenly by plunging into either cold water or oil. It makes steel hard and brittle.

(ii) ***Annealing :*** The steel is heated to red hot temperature (700 to 800°C) and then cooled slowly. It makes steel soft.

(iii) ***Tempering :*** If quenched steel is heated to temperature between 500 to 575 K and then cooled slowly, it becomes quite hard but brittleness disappears. The process is called tempering.

Surface treatment of steel

(i) ***Nitriding*** : Process of heating steel at 1000 K in an atmosphere of NH_3.
This gives hard coating of iron nitride on the surface.

(ii) ***Case hardening :*** Process of giving a thin coating of hardened steel, by heating steel in contact with charcoal followed quenching in oil.
– It is used for axles of railway wagons.

Uses

(i) ***Cast iron :*** It is the most important form of iron, is used for casting stoves, railway sleepers, gutter pipes , toys, etc.

(ii) It is used in the manufacture of wrought iron and steel.

(iii) ***Wrought iron :*** It is used in making anchors, wires, bolts, chains and agricultural implements.

(iv) ***Steel finds a number of uses :*** Alloy steel is obtained when other metals are added to it. Nickel steel is used for making cables, automobiles and aeroplane parts, pendulum, measuring tapes, chrome steel for cutting tools and crushing machines, and stainless steel for cycles, automobiles, utensils, pens, etc.

❑ EXTRACTION OF SILVER & GOLD

(1) Extraction of silver :

Occurrence :

Ag found in free and combined state in nature.

Its main ore is Argentite Ag_2S.

Other ores are

Copper silver glance – Cu_2S . Ag_2S

Horn silver – AgCl

Argentiferrous lead – PbS(0.01 – 0.1% Ag)

Steps of Metallurgy :

(A) From Argentite (Ag_2S)

(i) Concentration : As it is a sulphide ore, so froth floatation process is used.

(B) Leaching and reduction (Mac Arther cyanide process)

(a) Formation of cyanide complex :

(a) $Ag_2S + 4NaCN \rightleftharpoons 2Na[Ag(CN)_2] + Na_2S \xrightarrow{+O_2+H_2O} Na_2SO_4 + 'S' + 2NaOH$

[$\because$ O_2 is used to make reaction irreversible which remove Na_2S as Na_2SO_4 + S]

Na_2SO_4 does not reacts with sod. Argento cyanide.

(b) Displacement of Ag metal :–

$2Na[Ag(CN)_2] + Zn \rightarrow Na_2[Zn(CN)_4] + 2Ag$

Sodium tetra cyanozincate (black ppt) .

$Ag \xrightarrow[KNO_3]{\Delta}$ Ag (Solid) Shining silver (impure)

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(C) Purification by electrolytic method :

Pure Ag — Cathode

Impure Ag — Anode

Electrolyte — $AgNO_3$

+ 1% HNO_3 — To increase ionisation and to avoid hydrolysis of $AgNO_3$ by common ion effect

(D) Ag from Argentiferrous lead (PbS – Galena) :

Galena has a little amount of silver

(1) Concentration : (PbS) by froth floatation process

(E) Reduction of metal :

(a) Parke's process : Based on distribution law.

(i) Molten Ag is more soluble in molten zinc.

Pb (molten) → Zn(molten)

(Ag) ↗

(ii) Melting point of Zn/Ag alloy is greater than lead, so Zn/Ag freezes first.

(iii) Density of Zn/Ag is less than Pb so Zn/Ag forms the upper layer and Pb form lower layer.

(iv) Upper layer taken out.

(v) Ag is separated by using distillation method. (Vapourisation of Zn)

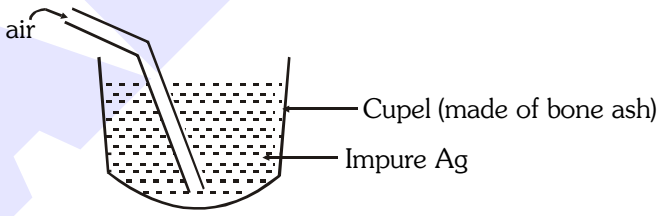
(b) Pattinson method :–

(i) It is based upon fractional crystallisation.

(ii) Ag/Pb is melted then cooled, the process is repeated again & again, concentration of Ag increases. Pb separate out in crystalline form.

(F) Purification :

(a) Cupellation :



Pb impurities separate (oxidised) out in the form of litharge.

(b) Electrolytic process :– As discussed earlier.

Uses : (i) It is used in silver plating.

(ii) Silver foils are used in medicine.

(iii) Silver amalgam is used for dental filling.

(iv) Compounds of silver are used in silvering of mirrors ($AgNO_3$ + HCHO + Red Pb), in photography, as laboratory reagents etc.

(v) Silver is easily alloyed with copper, so it is used in making coins, ornaments, silver ware etc.

(vi) It gives black spot on skin due to decomposition so it is also used as hair dye and ink.

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Extraction of Gold :

Occurrence : Au found in free (native) state in nature.

Same steps as are involved in the extraction of Silver metal.

(i) Crushing & Grinding :

(ii) Leaching process : Gold are extracted by the cyanide process (Mc Arthur - Forest process).

Reaction involved :

$$4Au + 8NaCN + O_2 + 2H_2O \rightleftharpoons 4Na[Au(CN)_2] + 4NaOH$$

$$2Na[Au(CN)_2] + Zn \longrightarrow Na_2[Zn(CN)_4] + 2Au$$

(iii) Refining Process :

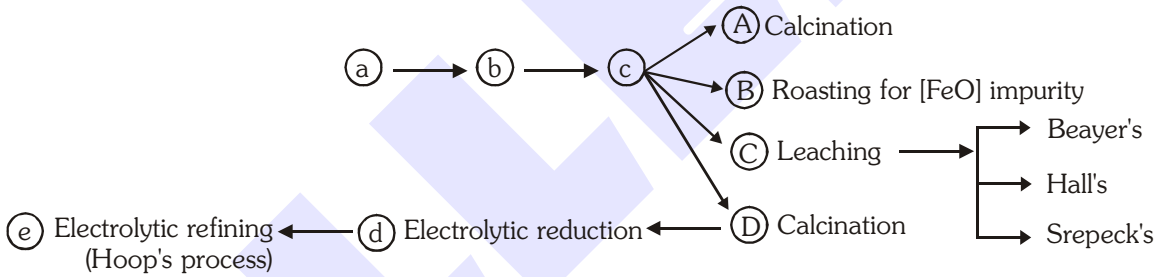
Anode	→	Impure Au
Cathode	→	Pure Au
Electrolyte	→	($AuCl_3$ solution + dil. HCl)

❑ EXTRACTION OF ALUMINIUM :

Ore - Bauxite $AlO_x(OH)_{3-2x}$ (where $O < x < 1$)

Extraction of Aluminium :

Short chart of Al from $Al_2O_3.2H_2O$ (Bauxite)



BAUXITE

↓

Step-1 : Concentration of Bauxite ore

(a) Baeyer's process : (Used for red bauxite in which main impurity is iron oxide)

Bauxite ore $\xrightarrow{\text{Roasted}}$ as to convert FeO into Fe_2O_3

Roasted ore + NaOH solution $\xrightarrow[8\text{ atm}]{150^\circ C}$ $NaAlO_2$ $\xrightarrow{\text{Hydrolysis}}$ $Al(OH)_3$ + NaOH in presence ppt. of little $Al(OH)_3$

(b) Hall's Process : (Red bauxite)

Bauxite ore + Na_2CO_3 $\xrightarrow{\text{Fused}}$ $NaAlO_2$

↓ extracted with water

Solution

↓ warmed 50° – 60° C

CO_2 is circulated

↓

$Al(OH)_3 + Na_2CO_3$

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(c) Serpeck's process : (Used for white bauxite in which main impurity is silica)

$$\text{Bauxite ore} + \text{coke} \xrightarrow{1800^\circ C} AlN \xrightarrow{H_2O} Al(OH)_3 + NH_3$$

$$+ \qquad\qquad\qquad\qquad\qquad\qquad \text{ppt.}$$

$$\text{(Nitrogen) } N_2 \qquad\qquad C + SiO_2 \text{ (gangue)} \rightarrow CO_2\uparrow + Si\uparrow$$

$$\downarrow$$

***Step-2:* CALCINATION**

$$Al(OH)_3 \xrightarrow{1500^\circ C} Al_2O_3$$

$$\downarrow$$

***Step-3 :* Electrolytic Reduction**

Electrolyte Al_2O_3 dissolved in Na_3AlF_6 and CaF_2

Cathode–Carbon lining

Anode – Graphite rods

Electrolytic reduction (Hall-Heroult process) :

The purified Al_2O_3 is mixed with Na_3AlF_6 (cryolite) or CaF_2 (fluorspar)) which lowers the melting point of the mixture and brings conductivity. The fused matrix is electrolysed. Steel cathode and graphite anode are used. The graphite anode is useful here for reduction to the metal. The overall reaction may be taken as :

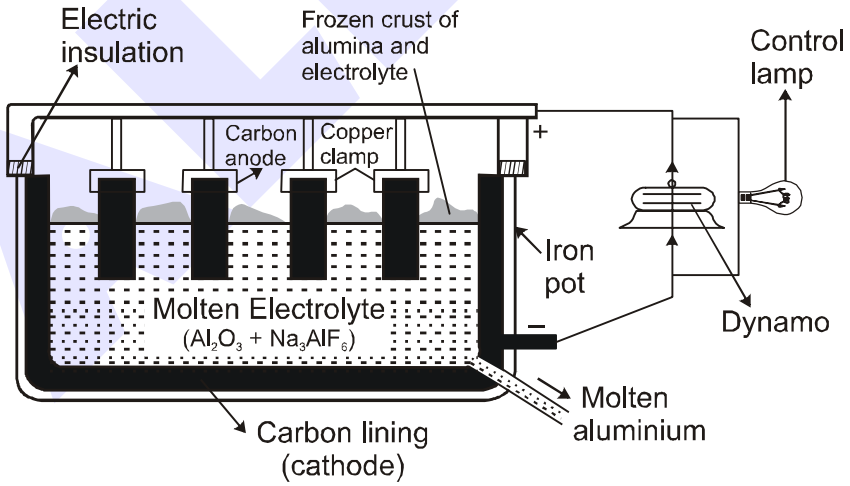
$$2Al_2O_3 + 3C \longrightarrow 4Al + 3CO_2$$

The electolysis of the molten mass is carried out in an electrolytic cell using carbon electrodes. The oxygen liberated at anode reacts with the carbon of anode producing CO and CO_2. This way for each kg of aluminium produced, about 0.5 kg of carbon anode is burnt away. The electrolytic reactions are :

Cathode : $Al^{3+}(melt) + 3e^- \longrightarrow Al(l)$

Anode : $C(s) + O^{2-}(melt) \longrightarrow CO(g) + 2e^-$

$C(s) + 2O^{2-}(melt) \longrightarrow CO_2(g) + 4e^-$

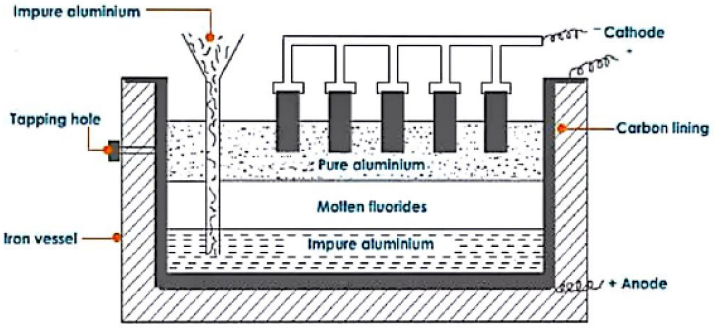


$$Al_2O_3 \xrightarrow[950^\circ C]{\text{Electrolysis}} Al + O_2$$

$\downarrow$ 99.8% pure

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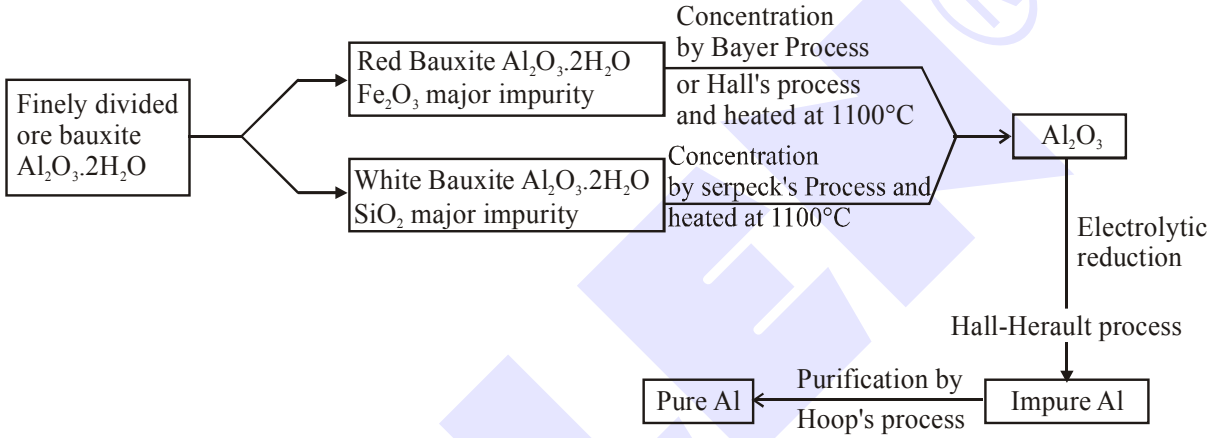
Step-4 : **Electrolytic Refining**



(Hoop's process)

Pure Al (99.98 % pure)

Flow chart of Al from $Al_2O_3.2H_2O$ (Bauxite)



Important points :

(a) Useful gas NH_3 is evolved in the leaching of bauxite by serpeck's process.

(b) In the electrolytic reduction of Al_2O_3 cryotite (Na_3AlF_6) is added along with CaF_2 (fluorspar) to–
– decrease m.p. of Al_2O_3
– decrease viscocity of electrolyte (CaF_2 is used)
– increase conductivity

(c) In the electrolytic reduction graphite anode get corrode or finishe due to reaction with O_2 liberates at anode, hence it had to be changed periodically.

(d) In the electrolytic refining (4th step) no electrodes are used. In the Hoop's process molten pure Al is used as cathode and molten impure Al is used as anode.

(e) In the Hoop's process carbon dust is sprayed over molten Al to
– avoid heat lose
– minimise metallic lusture (glaze) which is harmful for the eyes.

Uses

(i) Aluminium foils are used as wrappers for chocolates.

(ii) The fine dust of the metal is used in paints and lacquers.

(iii) Aluminium, being highly reactive, is also used in the extraction of chromium and manganese from their oxides.

(iv) Wires of aluminium are used as electricity conductors.

(v) Alloys containing aluminium, being light, are very useful.

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❑ EXTRACTION OF MAGNESIUM :

(1) Extraction of Magnesium :

(i) From Carnallite : Carnallite $KCl.MgCl_2.6H_2O$

The ore is dehydrated in a current of hydrogen chloride and the mixture of fused chlorides is electrolysed.

(ii) From Magnesite :

The concentrated ore is calcined at higher temperature

$$MgCO_3 \xrightarrow[\text{Strongly}]{\text{Heated}} MgO + CO_2$$

The calcined ore is heated with coke in a current of dry chlorine gas.

$$MgO + C + Cl_2 \xrightarrow{\Delta} MgCl_2 + CO$$

The magnesium chloride is fused and then electrolysed.

$$MgO + C \text{ (Other reducing agents like Si, Al can be used)} \xrightarrow[\text{Vaccume}]{2000°C} Mg + CO$$

(iii) From Sea water (Dow's process) :

Sea water contains 0.13% magnesium as chloride and sulphate. It involves following steps.

(a) *Precipitation of magnesium as magnesium hydroxide by slaked lime :*

$$MgCl_2 + Ca(OH)_2 \longrightarrow Mg(OH)_2 \downarrow + CaCl_2$$

(b) *Preparation of hexahydrated magnesium chloride*

$$Mg(OH)_2 + 2HCl(aq) \longrightarrow MgCl_2 + 2H_2O$$

The solution on concentration and crystallisation gives the crystals of $MgCl_2.6H_2O$

(c) *Preparation of anhydrous magnesium chloride*

$$MgCl_2.6H_2O \xrightarrow[\text{Dry HCl(g)}]{\Delta\text{(calcination)}} MgCl_2 + 6H_2O$$

It is not made anhydrous by simple heating because it gets hydrolysed

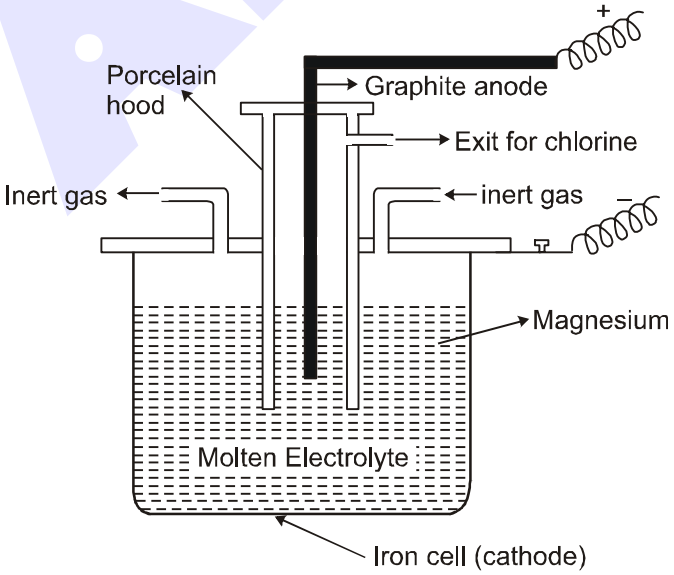
$$MgCl_2.\,6H_2O \xrightarrow{\Delta} MgO + 5H_2O + 2HCl$$

(d) *Electrolysis of fused anhydrous $MgCl_2$*

(i) Electrolyte : Molten $MgCl_2 + NaCl + CaCl_2$

(ii) Anode : Graphite electrode

(iii) Cathode : Iron cell (steel container)



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Magnesium chloride obtained by any of the above methods is fused and mixed with sodium chloride and calcium chloride in the temperature range of 973 – 1023 K. The molten mixture is electrolysed. Magnesium is liberated at the cathode (iron pot) and chlorine is evolved at graphite anode.

Reaction occurs :

The molten mixture is electrolysed. Magnesium is liberated at the cathode (iron pot) and chlorine is evolved at graphite anode.

$MgCl_2 \rightleftharpoons Mg^{2+} + 2Cl^-$

At cathode : $Mg^{2+} + 2e^- \longrightarrow Mg(99\%\ pure)$;

At anode : $2Cl^- \longrightarrow Cl_2 + 2e^-$

A stream of coal gas is passed through the pot to prevent oxidation of magnesium metal. The magnesium obtained in liquid state is purified by distillation under reduced pressure. (1 mm of Hg at 873 K).

(iv) From Dolomite : In the Pidgeon Process Mg is Produced.

The concentrated ore is calcined at higher temperature

$CaCO_3 . MgCO_3 \xrightarrow{\Delta} CaO. MgO + 2CO_2$

It is then reduced by ferrosilicon at 1273 K under reduced pressure.

$2CaO. MgO + Fe–Si \xrightarrow{1150°C} 2Mg\uparrow + Fe + Ca_2SiO_4$

Extraction of Na :

The fused mixture of NaCl and $CaCl_2$ is taken in Down's cell which consists of circular iron cathode and carbon anode. On passing the electric current the following reactions take place :

Ionisation of NaCl : $NaCl \rightleftharpoons Na^+ + Cl^-$

Collection of Na at cathode : $Na^+ + e^- \rightarrow Na(\text{Reduction})$.

Collection of Cl_2 at anode : $Cl^- + e^- \rightarrow Cl\,(\text{Oxidation}),\ Cl + Cl \rightarrow Cl_2$.

Aluminium	1. Bauxite, Al_2O_3. x H_2O 2. Cryolite, Na_3AlF_6	Electrolysis of Al_2O_3 dissolved in molten Na_3AlF_6	For the extraction, a good source of electricity is required.
Iron	1. Haematite, Fe_2O_3 2. Magnetite, Fe_3O_4	Reduction of the oxide with CO and coke in Blast furnace	Temperature approaching 2170 K is required.
Copper	1. Copper pyrites, $CuFeS_2$ 2. Copper glance, Cu_2S 3. Malachite, $CuCO_3.Cu(OH)_2$ 4. Cuprite, Cu_2O	Roasting of sulphide partially and reduction	It is self reduction in a specially designed converter. The reduction takes place easily. Sulphuric acid leaching is also used in hydrometallurgy from low grade ores.
Zinc	1. Zinc blende or Sphalerite, ZnS 2. Calamine, $ZnCO_3$ 3. Zincite, ZnO	Roasting followed by reduction with coke	The metal may be purified by fractional distillation.

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